

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1: AI Model Design Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA0101  
AND EXPERT SYSTEMS

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### COURSE OUTCOME COVERED

**CO4:** Develop intelligent software agents using suitable agent architectures and learning techniques.(BTL 3)

Assessment Tool Weightage: 40 %

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**Total Marks: 100**

**Time: 90 Mins**

**No. of Questions: 1**

### Annexure A AI Model Design Exercise

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#### Code of Conduct:

*I M. Veera Naga Sai (192425253) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student: M. Veera Naga Sai**

| Criterion                                     | Weightage | Marks Obtained |
|-----------------------------------------------|-----------|----------------|
| 1. Problem Analysis and Domain Understanding  | 10%       |                |
| 2. Dataset Selection and Data Understanding   | 10%       |                |
| 3. Data Preprocessing and Feature Engineering | 10%       |                |
| 4. AI Technique Selection and Justification   | 10%       |                |
| 5. Model Design / Architecture                | 10%       |                |
| 6. Algorithm / Pseudocode Development         | 5%        |                |
| 7. Model Implementation and GitHub Repository | 15%       |                |
| 8. Experimental Design and Results            | 10%       |                |
| 9. Performance Evaluation and Metrics         | 10%       |                |
| 10. Result Analysis and Interpretation        | 5%        |                |
| 11. Limitations and Future Enhancements       | 5%        |                |
| 12. Documentation and Technical Presentation  | 10%       |                |
| Total                                         | 100%      |                |

Signature of the Faculty

Total Marks Awarded

### ANSWER ANY ONE QUESTION

| S.No. | Scenario / Problem Statement                                                                                                                                                                                                               | AI Model / Technique to be Developed | Expected Development Outcome                                                                                                                                         |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <b>Smart Loan Approval:</b><br><br>A bank wants to automatically determine whether a customer is eligible for a loan using income, credit score, employment status, loan amount, and repayment history.                                    | Decision Tree                        | Design and implement a decision-tree-based AI model and determine the most suitable attribute-selection measure such as Information Gain, Gain Ratio, or Gini Index. |
| 2     | <b>Student Performance Prediction:</b><br><br>A university wants to predict whether a student is likely to pass or fail a course using attendance, internal marks, assignment performance, study hours, and previous academic performance. | Inductive Learning                   | Construct a predictive model from training examples and derive generalized decision rules from the learned patterns.                                                 |
| 3     | <b>Disease Risk Prediction:</b><br><br>A healthcare system needs to classify patients into low-risk and high-risk categories based on clinical measurements.                                                                               | Statistical Learning                 | Design a statistical learning model, train it using patient data, and evaluate its generalization ability on unseen data.                                            |
| 4     | <b>Image Recognition System:</b><br><br>An organization wants to automatically classify images into categories such as vehicles, animals, and buildings.                                                                                   | Neural Network                       | Design and implement a neural-network classifier by specifying the network architecture, activation functions, loss function, optimizer, and training procedure.     |
| 5     | <b>Handwritten Character Recognition:</b><br><br>A system must recognize handwritten characters from                                                                                                                                       | Multilayer Neural Network            | Develop a multilayer neural network and demonstrate how backpropagation updates weights to minimize classification error.                                            |

|    |                                                                                                                                                                          |                             |                                                                                                                                                                                                                                                       |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | scanned images.                                                                                                                                                          |                             |                                                                                                                                                                                                                                                       |
| 6  | <b>Non-Linear Classification:</b><br><br>A dataset contains two classes that cannot be separated using a straight-line decision boundary.                                | SVM with Kernel Method      | Design an SVM-based classifier using an appropriate kernel function and justify the selected kernel based on the characteristics of the dataset.                                                                                                      |
| 7  | <b>Intelligent Recommendation Agent:</b><br><br>An online platform wants an agent that learns which recommendations to provide based on user responses.                  | Reinforcement Learning      | Design an RL model by defining the states, actions, rewards, policy, and learning strategy, and demonstrate how the agent learns from user feedback.                                                                                                  |
| 8  | <b>Autonomous Game Agent:</b><br><br>An AI agent must learn to play a game where successful actions receive positive rewards and unsuccessful actions receive penalties. | Reinforcement Learning      | Develop an RL-based game agent and demonstrate how its policy and performance improve through repeated interaction with the environment.                                                                                                              |
| 9  | <b>Explainable Diagnosis System:</b><br><br>A medical AI system predicts a disease but must also provide an explanation for its prediction.                              | Explanation-Based Learning  | Design an explanation-based learning approach using domain knowledge and training examples to generate an understandable justification for the prediction.                                                                                            |
| 10 | <b>AI Model Selection Challenge:</b><br><br>A company provides a dataset for predicting customer churn.                                                                  | Comparative AI Model Design | Design and compare at least two learning approaches from decision trees, statistical learning, neural networks, or kernel methods. Select the best model based on accuracy, interpretability, computational requirements, and generalization ability. |

## Structure of the Report:

| S.No. | Report Component                               | What the Student Should Include                                                                                                          |
|-------|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <b>Problem Statement</b>                       | Clearly describe the real-world problem, input data, expected output, and AI task to be solved.                                          |
| 2     | <b>Objectives</b>                              | State the specific goals of developing the AI model and expected outcomes.                                                               |
| 3     | <b>Dataset Description</b>                     | Provide dataset source, number of instances, features, target variable, data types, and important attributes.                            |
| 4     | <b>Data Preprocessing</b>                      | Explain data cleaning, missing-value handling, encoding, normalization/scaling, feature selection, and train-test splitting.             |
| 5     | <b>Selected AI Technique and Justification</b> | Identify the selected AI technique and justify its suitability for the problem.                                                          |
| 6     | <b>Model Design / Architecture</b>             | Present the model structure, components, input/output, parameters, hyperparameters, and architecture/flow diagram.                       |
| 7     | <b>Algorithm / Pseudocode</b>                  | Provide the algorithmic steps or pseudocode showing model training and prediction/decision-making.                                       |
| 8     | <b>Implementation</b>                          | Provide programming language, libraries/tools, important code sections, and implementation details.                                      |
| 9     | <b>Experimental Results</b>                    | Present results using tables, graphs, sample predictions, learning curves, confusion matrices, or other relevant outputs.                |
| 10    | <b>Performance Evaluation</b>                  | Evaluate the model using appropriate performance metrics such as accuracy, precision, recall, F1-score, MSE, RMSE, or cumulative reward. |
| 11    | <b>Result Analysis</b>                         | Interpret results, identify important patterns, compare actual and predicted outcomes, and discuss strengths and weaknesses.             |
| 12    | <b>Limitations</b>                             | Discuss limitations related to dataset, model complexity, computational resources, accuracy, generalization, or deployment.              |
| 13    | <b>Future Enhancements</b>                     | Suggest improvements such as additional data, feature engineering, hyperparameter tuning, alternative algorithms, or deployment.         |
| 14    | <b>Conclusion</b>                              | Summarize the problem, methodology, major findings, model performance, and overall effectiveness.                                        |
| 15    | <b>References</b>                              | List datasets, research papers, books, websites, documentation, libraries,                                                               |

|    |                               |                                                                                                                                                                  |
|----|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                               | and other sources used.                                                                                                                                          |
| 16 | <b>GitHub Repository Link</b> | Provide the public GitHub repository URL containing the complete source code, dataset information/source, implementation, results, README, and supporting files. |

# **Disease Risk Prediction Using Statistical Learning**

## **1. Problem Statement**

Healthcare organizations need to identify patients who may have a high risk of developing a disease based on their clinical measurements. Manual analysis of multiple patient parameters can be time-consuming and may result in inconsistent decisions.

The proposed system uses a statistical learning model to classify patients into two categories: Low Risk and High Risk. The input data consists of clinical measurements such as age, blood pressure, cholesterol level, blood glucose level, body mass index (BMI), heart rate, and other relevant health attributes. The expected output is a risk classification for each patient.

The AI task is a binary classification problem, where the model learns patterns from previously labeled patient data and predicts the disease-risk category of unseen patients.

## **2. Objectives**

The main objectives of the Disease Risk Prediction system are:

1. To collect and prepare clinical patient data for statistical analysis.
2. To identify important clinical features associated with disease risk.
3. To develop a statistical learning model for binary classification.
4. To train the model using historical patient data.
5. To evaluate the model on unseen test data.
6. To measure the model using accuracy, precision, recall, and F1-score.
7. To determine whether the model can generalize effectively to new patients.
8. To provide a simple and reliable prediction of low-risk or high-risk categories.

The expected outcome is a model that can correctly classify new patients while maintaining good generalization performance.

### 3. Dataset Description

For implementation, a suitable public healthcare dataset such as the UCI Heart Disease Dataset can be used. The dataset contains clinical information about patients and a target indicating the presence or absence of heart disease.

#### Dataset Information

- Dataset Source: UCI Machine Learning Repository
- Dataset Name: Heart Disease Dataset
- Number of Instances: Approximately 303 patient records
- Number of Main Features: 13 clinical features
- Target Variable: Heart disease presence/absence
- Problem Type: Binary Classification
- Data Type: Numerical and categorical data

#### Important Features

| Feature                 | Description                         |
|-------------------------|-------------------------------------|
| Age                     | Age of the patient                  |
| Sex                     | Gender of the patient               |
| Chest Pain Type         | Type of chest pain                  |
| Resting Blood Pressure  | Blood pressure at rest              |
| Cholesterol             | Serum cholesterol level             |
| Fasting Blood Sugar     | Blood sugar measurement             |
| Resting ECG             | Resting electrocardiographic result |
| Maximum Heart Rate      | Maximum heart rate achieved         |
| Exercise-Induced Angina | Angina caused by exercise           |
| ST Depression           | ST depression induced by exercise   |
| Slope                   | Slope of peak exercise ST segment   |
| Number of Major Vessels | Number of major vessels observed    |
| Thalassemia             | Thalassemia-related measurement     |
| Target                  | Disease-risk classification         |

For the proposed system, the original target can be converted into two classes:

- 0 – Low Risk
- 1 – High Risk

## 4. Data Preprocessing

Data preprocessing is performed before training the statistical learning model.

### Step 1: Data Cleaning

Duplicate records and invalid values are identified and removed. Data types are checked to ensure that numerical and categorical features are correctly represented.

### Step 2: Missing-Value Handling

Missing clinical values can be handled using suitable methods such as:

- Mean or median imputation for numerical features.
- Mode imputation for categorical features.
- Removal of records when a large amount of information is missing.

Median imputation is preferred for numerical medical measurements because it is less affected by extreme values.

### Step 3: Encoding

Categorical variables are converted into numerical form using techniques such as:

- Label encoding for binary variables.
- One-hot encoding for multi-category variables.

### Step 4: Feature Scaling

Numerical features may have different ranges. For example, age and cholesterol can have different numerical scales.

Standardization can be applied using:

Standardized Value =  $(X - \text{Mean}) / \text{Standard Deviation}$

This ensures that features have comparable scales.

### Step 5: Feature Selection

Important clinical features are selected based on statistical relevance, correlation analysis, domain knowledge, or feature-importance methods.

### Step 6: Train-Test Splitting

The dataset is divided into:

- 80% Training Data
- 20% Testing Data

The training data is used to learn the model, while the test data is kept unseen during training to evaluate generalization ability.



## 5. Selected AI Technique and Justification

Selected Technique: Logistic Regression

Logistic Regression is selected as the statistical learning technique for disease-risk classification.

Logistic Regression is suitable because the expected output contains two categories: Low Risk and High Risk.

The model estimates the probability that a patient belongs to the high-risk class.

The logistic function is:

$$P(Y = 1) = 1 / (1 + e^{-z})$$

where:

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

The predicted probability is converted into a class using a threshold, commonly 0.5.

Reasons for Selection

1. Suitable for binary classification.
2. Simple and computationally efficient.
3. Easy to interpret.
4. Provides probability-based predictions.
5. Works well with numerical clinical measurements.
6. Coefficients can provide information about the influence of features.
7. Suitable as a baseline statistical learning model.
8. Can generalize to unseen data when properly trained and validated.

## 6. Model Design / Architecture

The proposed model follows a simple statistical learning pipeline.

Model Components

Input Layer

Patient clinical measurements such as:

- Age
- Blood pressure
- Cholesterol
- Heart rate
- Blood sugar
- Chest pain type

- Other clinical attributes



## Data Preprocessing

- Cleaning
- Missing-value handling
- Encoding
- Feature scaling
- Feature selection



## Training Dataset

80% of patient records



## Logistic Regression Model

The model learns the relationship between clinical measurements and disease risk.



## Probability Prediction

The model generates a probability between 0 and 1.



## Decision Threshold

- Probability  $\geq 0.5 \rightarrow$  High Risk
- Probability  $< 0.5 \rightarrow$  Low Risk



## Output

Low Risk / High Risk

## Architecture Flow Diagram

Patient Clinical Data



Data Cleaning



Missing Value Handling



Encoding & Feature Scaling



Feature Selection

↓

Train-Test Split

↓

Logistic Regression Training

↓

Probability Estimation

↓

Classification Threshold

↓

Low Risk / High Risk

## **7. Algorithm / Pseudocode**

Algorithm: Disease Risk Prediction

Step 1: Start.

Step 2: Load the patient dataset.

Step 3: Identify input features and target variable.

Step 4: Clean the dataset and handle missing values.

Step 5: Convert categorical attributes into numerical representation.

Step 6: Normalize or standardize numerical features.

Step 7: Select important clinical features.

Step 8: Divide the dataset into training and testing sets.

Step 9: Initialize the Logistic Regression model.

Step 10: Train the model using the training dataset.

Step 11: Provide unseen test data to the trained model.

Step 12: Calculate the probability of high disease risk.

Step 13: Apply the classification threshold.

Step 14: If probability  $\geq 0.5$ , classify the patient as High Risk.

Step 15: Otherwise, classify the patient as Low Risk.

Step 16: Calculate evaluation metrics.

Step 17: Analyze the results.

Step 18: Stop.

## 8. Implementation

The system can be implemented using Python.

Programming Language

Python

Libraries and Tools

- Pandas – Dataset loading and manipulation
- NumPy – Numerical calculations
- Scikit-learn – Machine learning and evaluation
- Matplotlib – Data visualization
- Seaborn – Statistical visualization

Important Implementation Steps

1. Import the required libraries.
2. Load the healthcare dataset.
3. Separate features and target.
4. Preprocess the clinical data.
5. Split the data into training and testing sets.
6. Create a Logistic Regression model.
7. Train the model.
8. Generate predictions for unseen data.
9. Calculate performance metrics.
10. Generate a confusion matrix and other visualizations.

Model Training

The Logistic Regression model learns coefficients for the selected clinical features. During prediction, these coefficients are used to calculate the probability of a patient belonging to the high-risk category.

## 9. Experimental Results

After training, the model is evaluated using the test dataset that was not used during model training.

The experimental results can be represented using the following table:

| Metric           | Result                        |
|------------------|-------------------------------|
| Training Samples | Approximately 80% of dataset  |
| Testing Samples  | Approximately 20% of dataset  |
| Accuracy         | To be obtained experimentally |
| Precision        | To be obtained experimentally |
| Recall           | To be obtained experimentally |
| F1-Score         | To be obtained experimentally |

### Sample Prediction

| Patient   | Predicted Probability | Predicted Class |
|-----------|-----------------------|-----------------|
| Patient 1 | 0.23                  | Low Risk        |
| Patient 2 | 0.78                  | High Risk       |
| Patient 3 | 0.41                  | Low Risk        |
| Patient 4 | 0.86                  | High Risk       |

*The above values are illustrative sample predictions. Actual values depend on the dataset split and model implementation.*

### Confusion Matrix

The confusion matrix contains four categories:

|                  | Predicted Low Risk | Predicted High Risk |
|------------------|--------------------|---------------------|
| Actual Low Risk  | True Negative      | False Positive      |
| Actual High Risk | False Negative     | True Positive       |

The confusion matrix helps determine which types of predictions the model gets right or wrong.

## 10. Performance Evaluation

The model is evaluated using classification metrics.

Accuracy

Accuracy represents the percentage of correctly classified patients.

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Precision

Precision indicates how many patients predicted as high risk are actually high risk.

$$\text{Precision} = TP / (TP + FP)$$

Recall

Recall measures how many actual high-risk patients are correctly identified.

$$\text{Recall} = TP / (TP + FN)$$

For healthcare risk prediction, recall is particularly important because failing to identify a high-risk patient can have serious consequences.

F1-Score

F1-score combines precision and recall.

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Generalization Evaluation

The model's generalization ability is evaluated by comparing training performance with testing performance.

If the training and testing results are reasonably similar, the model is likely generalizing well.

A large difference between training and testing performance may indicate overfitting.

## 11. Result Analysis

The Logistic Regression model provides a probability-based prediction for each patient.

Clinical measurements are used to estimate the probability of belonging to the high-risk category.

A patient with a higher predicted probability is more likely to be classified as high risk.

Features such as age, blood pressure, cholesterol, chest pain type, and maximum heart rate can contribute to the classification.

The model has several advantages:

- It is simple and easy to understand.
- It provides probability-based predictions.
- It is computationally efficient.

- Its coefficients can be interpreted.
- It provides a strong baseline for medical risk classification.

However, the model may not capture complex nonlinear relationships between clinical variables. Therefore, advanced models may be considered if higher predictive performance is required.

The comparison between training and testing performance is also important. If both results are similar, the model demonstrates better generalization to unseen patients.

## **12. Limitations**

The proposed system has several limitations:

1. The dataset may contain a relatively small number of patient records.
2. Dataset quality directly affects prediction quality.
3. Missing or incorrect clinical measurements can reduce model performance.
4. Logistic Regression assumes a relatively simple relationship between input variables and the log-odds of the outcome.
5. Complex interactions between medical features may not be fully captured.
6. Results obtained from one dataset may not generalize to every hospital or population.
7. The model should not be treated as a replacement for professional medical diagnosis.
8. Real-world deployment requires extensive clinical validation and appropriate privacy and security measures.

## **13. Future Enhancements**

The system can be improved in several ways:

1. Use a larger and more diverse healthcare dataset.
2. Include additional clinical measurements.
3. Apply advanced feature-engineering techniques.
4. Perform hyperparameter optimization.
5. Compare Logistic Regression with Decision Tree, Random Forest, Support Vector Machine, and Neural Network models.
6. Apply cross-validation for more reliable performance estimation.
7. Handle class imbalance using appropriate techniques.
8. Develop an interactive web-based prediction interface.
9. Add explainable AI techniques to show why a patient is classified as high risk.

10. Deploy the model as a healthcare decision-support system after proper clinical validation.
11. Continuously monitor model performance after deployment.
12. Apply privacy-preserving techniques to protect patient information.

## **14. Conclusion**

The Disease Risk Prediction system demonstrates how statistical learning can be applied to healthcare classification problems. Clinical measurements are collected, cleaned, transformed, and divided into training and testing datasets. Logistic Regression is then trained to classify patients into low-risk and high-risk categories.

The model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

Testing the model on unseen data provides an estimate of its generalization ability.

Logistic Regression is an appropriate baseline because it is simple, interpretable, computationally efficient, and suitable for binary classification. However, additional datasets, feature engineering, cross-validation, and alternative machine learning techniques can be used to improve the system.

Overall, the proposed system can serve as a decision-support tool for disease-risk assessment, but actual medical decisions should be made by qualified healthcare professionals.

## **15. References**

1. UCI Machine Learning Repository, Heart Disease Dataset.
2. Scikit-learn Documentation, Logistic Regression and Classification Metrics.
3. Hastie, T., Tibshirani, R., and Friedman, J., The Elements of Statistical Learning, Springer.
4. James, G., Witten, D., Hastie, T., and Tibshirani, R., An Introduction to Statistical Learning, Springer.
5. Pedregosa, F. et al., Scikit-learn: Machine Learning in Python, Journal of Machine Learning Research.
6. Python Documentation, Python Programming Language.
7. Pandas Documentation, Data Analysis and Manipulation Using Python.
8. NumPy Documentation, Numerical Computing with Python.



## 16. GitHub Repository Link

The complete project can be uploaded to a public GitHub repository containing:

- Python source code
- Dataset information and source
- Data preprocessing procedures
- Model training implementation
- Model evaluation
- Confusion matrix
- Result graphs
- Sample predictions
- README file
- Requirements file
- Project documentation

GitHub Repository:

<https://github.com/your-username/disease-risk-prediction>

Note: Replace your-username with the actual GitHub username and repository name after uploading the project.

Suggested Repository Structure

disease-risk-prediction/

```
|
|— dataset/
|   └─ dataset_information.txt
|
|— src/
|   └─ disease_risk_prediction.py
|
|— results/
|   └─ confusion_matrix.png
|   └─ performance_results.txt
|
|— README.md
|— requirements.txt
└─ LICENSE
```

## Evaluation Rubrics:

| Criterion                                     | Weightage (%) | Excellent                                                                                                                                 | Good                                                                      | Average                                                                  | Poor                                                           |
|-----------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------|
| 1. Problem Analysis and Understanding         | 10%           | 9–10% – Clearly analyzes the real-world problem, objectives, inputs, outputs, constraints, and expected AI outcome.                       | 7–8% – Understands the problem and objectives with minor omissions.       | 5–6% – Shows partial understanding with some important elements missing. | 0–4% – Problem is poorly understood or incorrectly defined.    |
| 2. Dataset Selection and Understanding        | 10%           | 9–10% – Selects a highly relevant dataset and clearly explains features, target, source, size, and characteristics.                       | 7–8% – Appropriate dataset with adequate description and minor omissions. | 5–6% – Dataset is usable but description or relevance is limited.        | 0–4% – Dataset is inappropriate, poorly described, or missing. |
| 3. Data Preprocessing and Feature Engineering | 10%           | 9–10% – Performs appropriate cleaning, transformation, encoding, scaling, feature selection, and data splitting with clear justification. | 7–8% – Performs most required preprocessing correctly with minor issues.  | 5–6% – Basic preprocessing is performed but important steps are missing. | 0–4% – Preprocessing is inadequate, incorrect, or absent.      |
| 4. AI Technique Selection and Justification   | 10%           | 9–10% – Selects the most appropriate AI technique and provides strong technical justification based on the problem and dataset.           | 7–8% – Appropriate technique selected with reasonable justification.      | 5–6% – Technique is acceptable but justification is limited.             | 0–4% – Inappropriate technique or no meaningful justification. |
| 5. Model Design / Architecture                | 10%           | 9–10% – Develops a clear, technically sound model architecture with appropriate components, parameters, and workflow.                     | 7–8% – Model is well designed with minor omissions.                       | 5–6% – Basic model design is presented but lacks technical depth.        | 0–4% – Model design is unclear, incomplete, or inappropriate.  |

|                                         |            |                                                                                                                                                      |                                                                                     |                                                                                    |                                                                                     |
|-----------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 6. Algorithm / Pseudocode               | <b>5%</b>  | 5% – Algorithm/pseudocode is complete, logically correct, structured, and clearly represents the model development process.                          | 4% – Mostly correct with minor omissions.                                           | 2–3% – Basic algorithm provided but contains gaps.                                 | 0–1% – Algorithm is missing or substantially incorrect.                             |
| 7. Implementation and GitHub Repository | <b>15%</b> | 13–15% – Fully functional implementation with clean, organized, reproducible code and a complete GitHub repository with README and supporting files. | 10–12% – Functional implementation and well-organized repository with minor issues. | 7–9% – Partially functional implementation or incomplete repository documentation. | 0–6% – Implementation is incomplete/non-functional or GitHub repository is missing. |
| 8. Experimental Design and Results      | <b>10%</b> | 9–10% – Conducts meaningful experiments with appropriate test cases and clearly presents results using tables, graphs, or visualizations.            | 7–8% – Appropriate experiments with clearly presented results.                      | 5–6% – Limited experiments or basic result presentation.                           | 0–4% – Insufficient, incorrect, or missing experimental results.                    |
| 9. Performance Evaluation               | <b>10%</b> | 9–10% – Uses appropriate evaluation metrics and provides comprehensive quantitative analysis of model performance.                                   | 7–8% – Uses suitable metrics and provides adequate evaluation.                      | 5–6% – Basic evaluation with limited metrics or analysis.                          | 0–4% – Inappropriate metrics or performance evaluation is missing.                  |

|                                              |            |                                                                                                                                |                                                                       |                                                           |                                                          |
|----------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------|
| 10. Result Analysis and Interpretation       | <b>5%</b>  | 5% – Provides insightful analysis of results, identifies patterns, errors, strengths, weaknesses, and explains model behavior. | 4% – Provides good interpretation with minor gaps.                    | 2–3% – Basic interpretation with limited analysis.        | 0–1% – Results are not meaningfully analyzed.            |
| 11. Limitations and Future Enhancements      | <b>5%</b>  | 5% – Clearly identifies realistic limitations and proposes technically relevant and feasible improvements.                     | 4% – Identifies relevant limitations and reasonable improvements.     | 2–3% – Provides basic limitations and future suggestions. | 0–1% – Missing, unrealistic, or poorly explained.        |
| 12. Documentation and Technical Presentation | <b>10%</b> | 9–10% – Report is comprehensive, well organized, technically accurate, professionally presented, and properly referenced.      | 7–8% – Well documented with minor presentation or referencing issues. | 5–6% – Adequate documentation with several gaps.          | 0–4% – Poorly documented, unclear, or incomplete report. |
| Total                                        | 100%       |                                                                                                                                |                                                                       |                                                           |                                                          |